

## Lecture - 8

### Gradients and Hessian of Quadratic & Linear fns.

#### Gradient of linear fns. -

$\vec{x} \in \mathbb{R}^n$  let  $f(x) = \vec{b}^T \vec{x}$  for some vector  $\vec{b} \in \mathbb{R}^n$

$$f(x) = \sum_{i=1}^n b_i x_i$$

$$\frac{\partial f(x)}{\partial x_k} = \frac{\partial}{\partial x_k} \sum_{i=1}^n b_i x_i = b_k$$

$$\boxed{\nabla \vec{b}^T \vec{x} = \vec{b}}$$

$$\boxed{\frac{\partial (ax)}{\partial x} = a}$$

#### \* Gradient of quadratic fns. -

$$f(x) = \vec{x}^T \vec{A} \vec{x}$$

[A is a symmetric matrix]

$$f(x) = \sum_{i=1}^n \sum_{j=1}^n A_{ij} x_i x_j$$

$$\frac{\partial f(x)}{\partial x_k} = \frac{\partial}{\partial x_k} \sum_i \sum_j A_{ij} x_i x_j$$

$$= \frac{\partial}{\partial x_k} \left[ \sum_{i \neq k} \sum_{j \neq k} A_{ij} x_i x_j + \sum_{i \neq k} A_{ik} x_i x_k + \sum_{j \neq k} A_{kj} x_k x_j + A_{kk} x_k^2 \right]$$

$$= \sum_{i \neq k} A_{ik} x_i + \sum_{j \neq k} A_{kj} x_j + 2A_{kk} x_k$$

$+ A_{kk} x_k \quad + A_{kk} x_k$

$$= \sum_{i=1}^n A_{ik} x_i + \sum_{j=1}^n A_{kj} x_j$$

$$= \sum_{i=1}^n A_{ik} x_i + \sum_{i=1}^n A_{ki} x_i$$

$\therefore A$  is symmetric

$$\rightarrow = 2 \sum_{i=1}^n A_{ki} x_i$$

$$\boxed{\nabla_x \vec{x}^T \vec{A} \vec{x} = 2 \vec{A} \vec{x}}$$

$$\equiv \boxed{\frac{\partial (ax^2)}{\partial x} = 2ax}$$

\* Hessian of Linear Fm: -

$$f(x) = b^T x$$

$$\nabla_x^2 f(x) = 0$$

\* Hessian of Quadratic Fm: -

$$f(x) = x^T A x$$

$$\frac{\partial^2 f(x)}{\partial x_k \partial x_l} = \frac{\partial}{\partial x_k} \left[ \frac{\partial f(x)}{\partial x_l} \right] = \frac{\partial}{\partial x_k} \left[ 2 \sum_{i=1}^n A_{li} x_i \right]$$

$$= 2 A_{lk} = 2 A_{kl}$$

$$\boxed{\nabla_x^2 x^T A x = 2 A}$$

$$\equiv \boxed{\frac{\partial^2 (ax^2)}{\partial x^2} = 2a}$$

- Least Squares:-  $A \in \mathbb{R}^{m \times n}$  (full rank)  
 $b \in \mathbb{R}^m$  s.t.  $b \notin \mathcal{R}(A)$   
 $\mathcal{R}(A) \rightarrow$  Range of  $A$ .

$$Ax = b$$

We will not be able to find  $x$  for which

$$Ax = b$$

Find a vector  $x$  s.t.  $Ax$  is as close as possible to  $b$

$$\begin{aligned} & \|Ax - b\|_2^2 \\ &= (Ax - b)^T (Ax - b) \\ &= [(Ax)^T - b^T] (Ax - b) \\ &= [(x^T A^T) - b^T] (Ax - b) \\ &= x^T A^T A x - x^T A^T b - b^T A x + b^T b \\ &= x^T A^T A x - (\underline{Ax})^T b - b^T \underline{Ax} + b^T b \\ &= x^T A^T A x - 2 b^T A x + b^T b \\ &\nabla_x (x^T A^T A x - 2 b^T A x + b^T b) \\ &= \nabla_x x^T A^T A x - \nabla_x 2 b^T A x \end{aligned}$$

$\|\cdot\| \rightarrow$  Norm

$\|\cdot\|_2 \rightarrow L_2$  norm.

$\|\cdot\|_2^2 \rightarrow$  Squared  $L_2$ -norm.

$$\Rightarrow \|x\|_2^2 = x^T x$$

$$[a^T b = b^T a]$$

$$= 2 A^T A x - 2 A^T b \longrightarrow 0 \quad \cup$$

$$A^T A x = A^T b$$

$$\hat{x} = (A^T A)^{-1} A^T b$$

Least Square  
Solution